



BMBA 1133 Introduction to Data Analytics

Introduction to Data Analytics

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Learning Outcomes



Define Data
Analytics

01.

02.

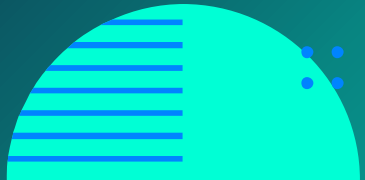
Understand
Different Data
Analysis Types

Explore Different
Data Analysis Tools

03.

04.

Demonstrate the
Typical Process of
a Business Analyst



Data Analytics

The process of collecting, inspecting, transforming and modelling data using various statistical and logical techniques and methods with the goal of discovering important information and supporting decision making

Why do we need Data Analysis



Retail

(Supermarkets analyze sales data to decide which products to promote)



Healthcare

(Hospitals study patient records to find the most effective treatments)



Sports

(Coaches use player stats to plan team strategies)

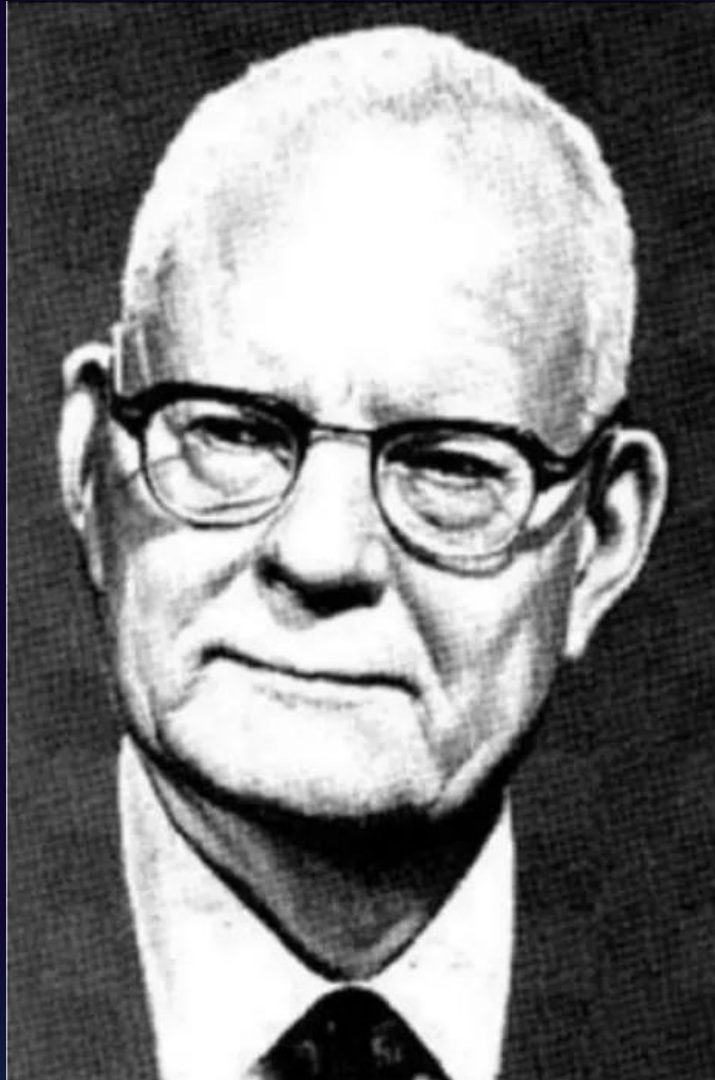


Location

(finding the best location for bank branches and ATMs)

Data analysis helps:

- ✓ Solve problems with evidence
 - ✓ Improve performance
- ✓ Predict future outcomes
- ✓ Make smarter decisions

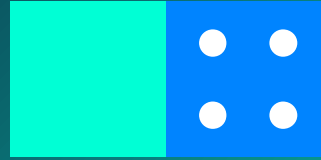


“Without data
you’re just
another person
with an opinion.”

- W. Edwards Deming,
Data Scientist



Four Main Pillars/Types of Data Analysis



Descriptive Data Analytics



Diagnostic Data Analytics



Predictive Data Analytics

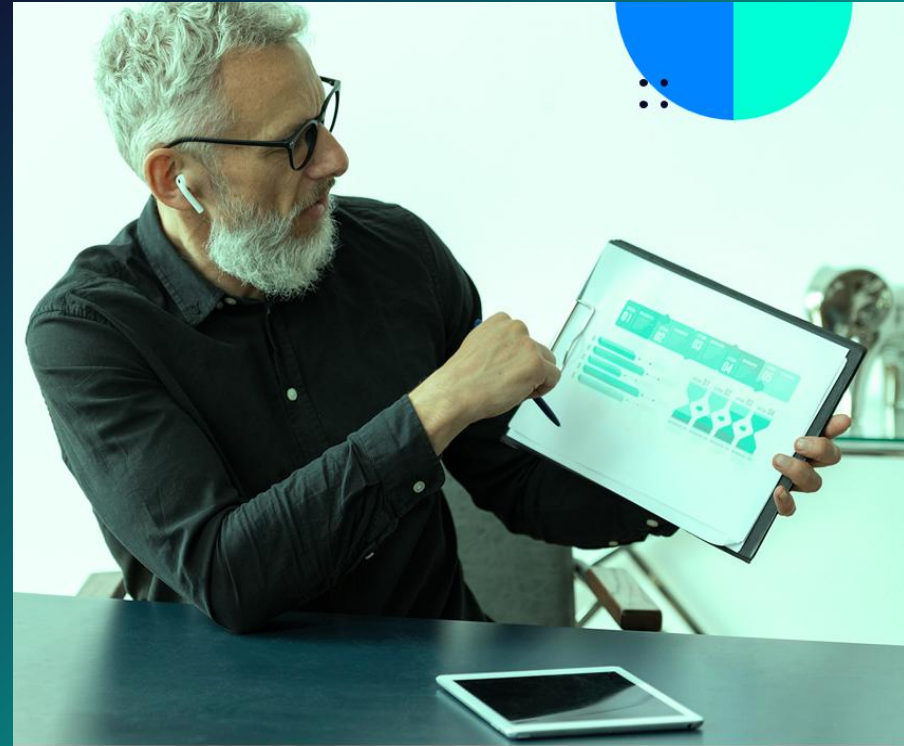


Prescriptive Data Analytics

Descriptive Data Analytics

The process of using historical data to identify what has happened in the past

Example: Financial statement analysis, traffic reports



Examples

1. Student Performance

A report showing average marks for each subject in the last semester

- *70% of students passed Legal Environment*
- *Average mark in Accounting for Business was 62*

2. Retail

A shop analyses last month's sales

- *We sold 300 units of Product A in June*
- *Revenue increased by 10% compared to May*

3. Website Visitors

A website tracks user activity

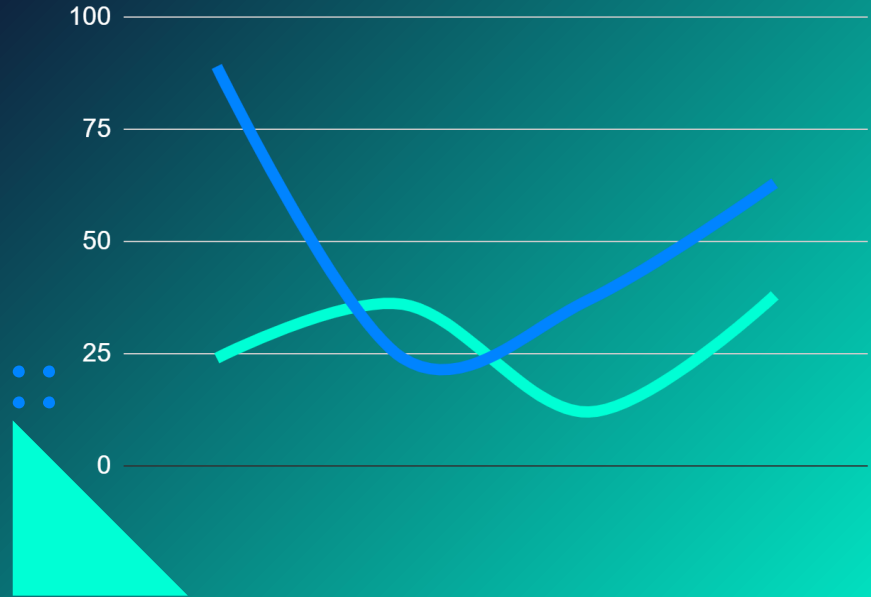
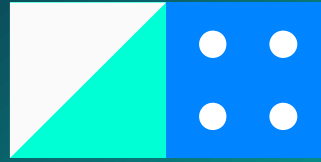
- *We had 5,000 visitors this week*
- *Most visitors accessed from mobile devices*



Diagnostic Data Analytics

The process of using data to examine 'why did it happen?'

Example: Reasons for last month sales reduction



Examples

1. Student Performance

After noticing a high failure rate, the university investigates further

- 🔍 *Most failing students had less than 50% attendance*
- 🔍 *Failure rate was higher in classes with larger classes*

2. Retail

A store finds that sales dropped in May and wants to know why

- 🔍 *Sales dropped due to a stockout of top products*
- 🔍 *Competitor launched a big promotion in the same month*

3. Website Traffic

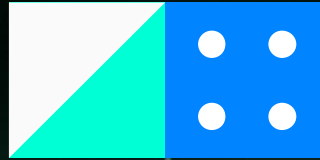
A website's visitors suddenly decreased last week

- 🔍 *Majority of the drop was from mobile users after a new update caused display issues on phones*

Predictive Data Analytics

The process of using data to
forecast future results

Example: Demand
forecast for future period



Examples

1. Student Performance

The university wants to predict which students are likely to fail a subject

 *Students with low attendance have a 75% of chance of failing*

 *Students with low midterm marks have a 45% of chance of failing*

2. Retail

A store uses past sales data to forecast next month's demand

 *Based on last 3 years of sales, we expect ice cream sales to increase in August*

 *Predicted demand for Product A is 1,200 units next week*

3. Website Traffic

A website owner predicts the risk of loosing customers

 *Mobile users experiencing display issues are likely to abandon the site, with a 70% probability of leaving before engaging with content*



Prescriptive Data Analytics

The process of using data to
identify an optimal course of
action

Example: Suggestions to
overcome market share drop

Recommends actions that can take
to affect the outcomes of predictive
analysis



Examples

1. Student Support

The university identifies at-risk students and recommends support

- *Based on past data and predictions, the system recommends weekly mentoring sessions for students at risk*
- *Schedule changes are suggested to improve student attendance*

2. Retail

A store wants to increase sales next month

- *The system suggests running a 20% discount on Product A during weekends, based on predicted demand and profit margins*
- *Adjust stock levels and staff shifts to match predicted high-demand periods*

3. Website Traffic

A website needs to regain customers

- *Optimize the mobile user interface for better engagement*
- *Send targeted notifications or promotions to re-engage affected users*

In Simple Terms:

Descriptive = It rained

Diagnostic = It rained because of low pressure

Predictive = It might rain tomorrow

Prescriptive = Take an umbrella or avoid stepping out of the house



Data Analysis Tools

BI Tools

Tableau
Power BI

01

Spreadsheet Software

Microsoft Excel
Google Sheets

02

Database Tools

SQL
Mongo DB

03

Data Analysis Tools Cont.

Programming Languages

Python
R

04

Statistical Tools

SPSS
SAS

05

Typical Process of a Data Analyst



DEFINE THE
QUESTION



COLLECT THE
DATA



CLEAN THE
DATA



ANALYZE THE
DATA



INTERPRET
AND SHARE
THE RESULT



Thank You

